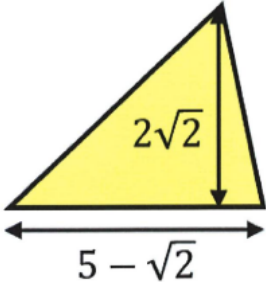
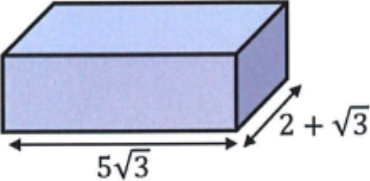
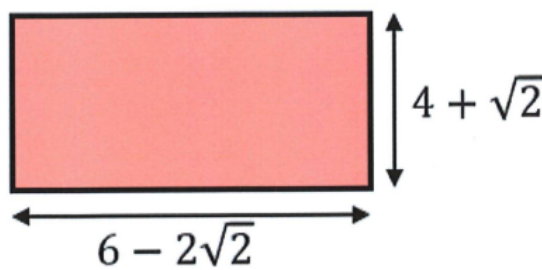


Surd Equations & Problem Solving Questions

1)	Given that $\frac{10 - \sqrt{32}}{\sqrt{2}} = a + b\sqrt{2}$ where a and b are integer. Find the values of a and b.	2)	Show that the following surd expression can be written as $\frac{a}{b}\sqrt{c}$, where a, b and c are all integers: $\left(\frac{4}{3}\right)^{\frac{1}{2}} + \left(\frac{1}{3}\right)^{-\frac{1}{2}}$
3)	$a = \sqrt{8} + 2$ $b = \sqrt{8} - 2$ $T = a^2 - b^2$ Work out the value of T. Give your answer in the form $c\sqrt{2}$ where c is an integer.	4)	Expand $(8 - \sqrt{3})^2$ giving your answer in form $a + b\sqrt{3}$
5)	The area of a rectangle is 18 cm^2 The length of the rectangle is $(\sqrt{7} + 1) \text{ cm}$. Without using a calculator and showing each stage of your working, find the width of the rectangle. Give your answer in the form $a\sqrt{b} + c$ where a, b and c are integers.	6)	Solve for n. $2^{4n} - \frac{1}{\sqrt[4]{16}} = 0$
7)	Express $x^2 + 6\sqrt{2}x - 1$ in the form $(x + a)^2 + b$ Show your working clearly.	8)	Find the area of the triangle. 
9a)	(a) The volume of the cuboid shown is $15 + 5\sqrt{3}$. Find the height of the cuboid. 	9b)	(b) The volume of a cylinder is $40\pi + 20\pi\sqrt{2}$. If the height of the cylinder is $2 + \sqrt{2}$, find the exact radius of the cylinder.

10)	Solve for x $x^2 - 2\sqrt{3}x + 3 = 0$	11)	The perimeter of this rectangle is $8\sqrt{2}$. Find the area of the rectangle. 
12)	Solve for x $3x - \sqrt{5} = \sqrt{5}x + 1$	13)	$(4 + \sqrt{a})(7 - \sqrt{a}) = 23 + k\sqrt{a}$ Find the values of the positive integers a and k.
14)	Find the area: 	15)	Rationalise the denominator of the expression $\frac{1+\sqrt{5}}{3+\sqrt{5}}$. Give your answer in its simplest form:
16)	Express $\frac{3+\sqrt{3}}{3(\frac{1}{\sqrt{3}})}$ in the form $a + \sqrt{b}$, where both a and b are integers.	17)	Simplify $\sqrt{80} - \frac{10}{\sqrt{5}}$
18)	Simplify the following expression: $\sqrt{4\frac{12}{9} + \left(\frac{1}{3}\right)^{\frac{1}{2}}}$	19)	

Answers

1)	$\begin{array}{cc} -4 + 5\sqrt{2} \\ \downarrow \quad \downarrow \\ a \quad b \end{array}$	2)	$\frac{5\sqrt{3}}{3} \quad \begin{array}{l} a=5 \\ b=3 \\ c=3 \end{array}$
3)	$16\sqrt{2}$	4)	$(67 - 16\sqrt{3})$
5)	$\begin{array}{l} 18 = (\sqrt{7} + 1)x \\ x = 3\sqrt{7} - 3 \end{array}$	6)	$h = -\frac{1}{4}$
7)	$(x + 3\sqrt{2})^2 - 19$	8)	$5\sqrt{2} - 2$
9a)	$\begin{array}{l} 15 + 5\sqrt{3} = (5\sqrt{3})(2 + \sqrt{3})h \\ h = \sqrt{3} - 1 \end{array}$	9b)	$\begin{array}{l} \text{d the exact radius} \\ 40\pi + 20\pi\sqrt{2} = \pi h^2(2 + \sqrt{2}) \\ h = 2\sqrt{5} \end{array}$
10)	$\begin{array}{l} (x - \sqrt{3})(x - \sqrt{3}) = 0 \\ x = \sqrt{3} \end{array}$	11)	$\begin{array}{l} 8\sqrt{2} = (\sqrt{2} - 1) + (\sqrt{2} - 1) + 2x \\ x = 3\sqrt{2} + 1 \\ A = (\sqrt{2} - 1)(3\sqrt{2} + 1) \\ = 5 - 2\sqrt{2} \end{array}$
12)	$\begin{array}{l} 3x - \sqrt{5}x = 1 + \sqrt{5} \\ x(3 - \sqrt{5}) = 1 + \sqrt{5} \\ x = \frac{1 + \sqrt{5}}{3 - \sqrt{5}} \\ x = 2 + \sqrt{5} \end{array}$	13)	$\begin{array}{l} a = 5 \\ k = 3 \end{array}$
14)	$\begin{array}{l} (4 + \sqrt{2})(6 - 2\sqrt{2}) \\ = 20 - 2\sqrt{2} \end{array}$	15)	$\frac{-1 + \sqrt{5}}{2}$
16)	$1 + \sqrt{3}$	17)	$2\sqrt{5}$
18)	$\frac{5\sqrt{3}}{3}$	19)	